

Mechanical design of a heliostat structure

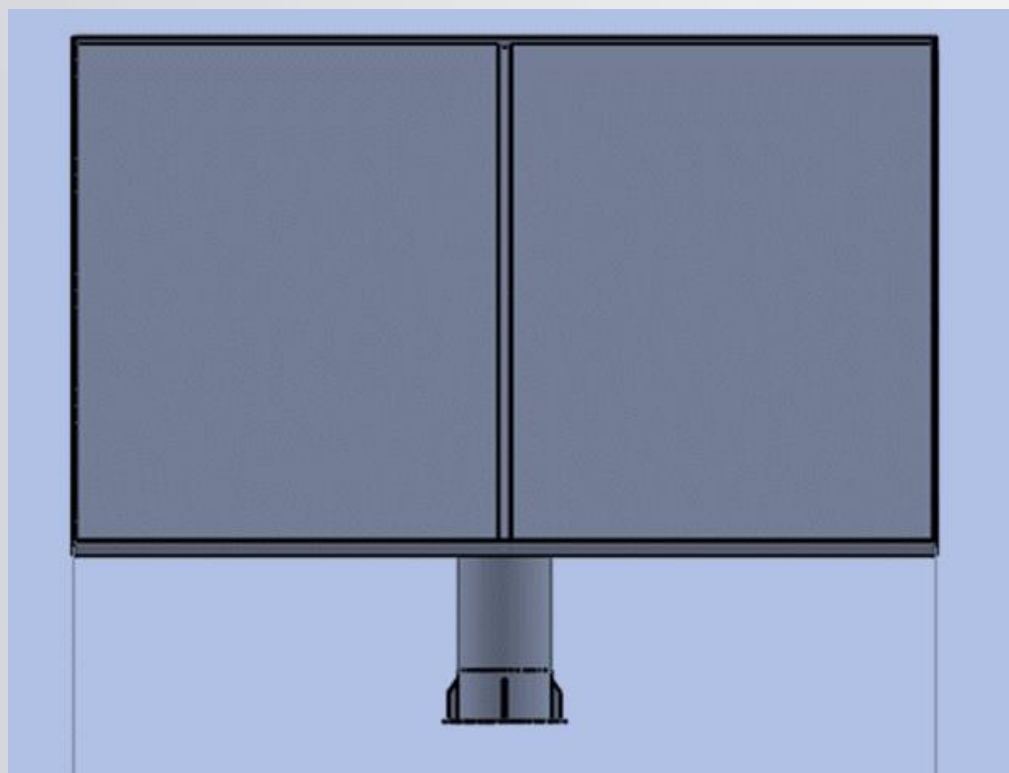
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Dr. Mehmet Guler

PROBLEM STATEMENT, CRITERIA AND CONSTRAINTS

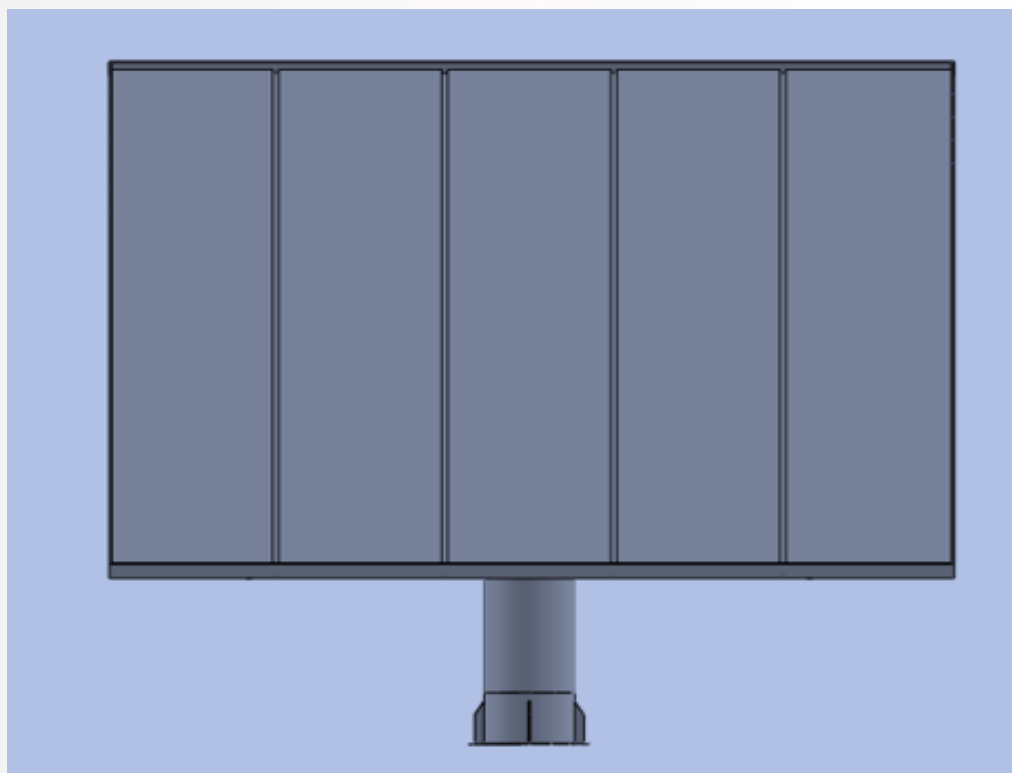
The challenge is to develop a heliostat design that can withstand wind loads while maintaining cost-effectiveness and ensuring sustainable safety.

- The Cell size should be 7 m × 7 m
- The column height should be 3.5 m
- The diameter of column should be not less than 1 m
- The material should be construction steel
- Stow velocity does not exceed 50 m/s.
- Orientation angle $\alpha = 0, 35, 70$ and 90

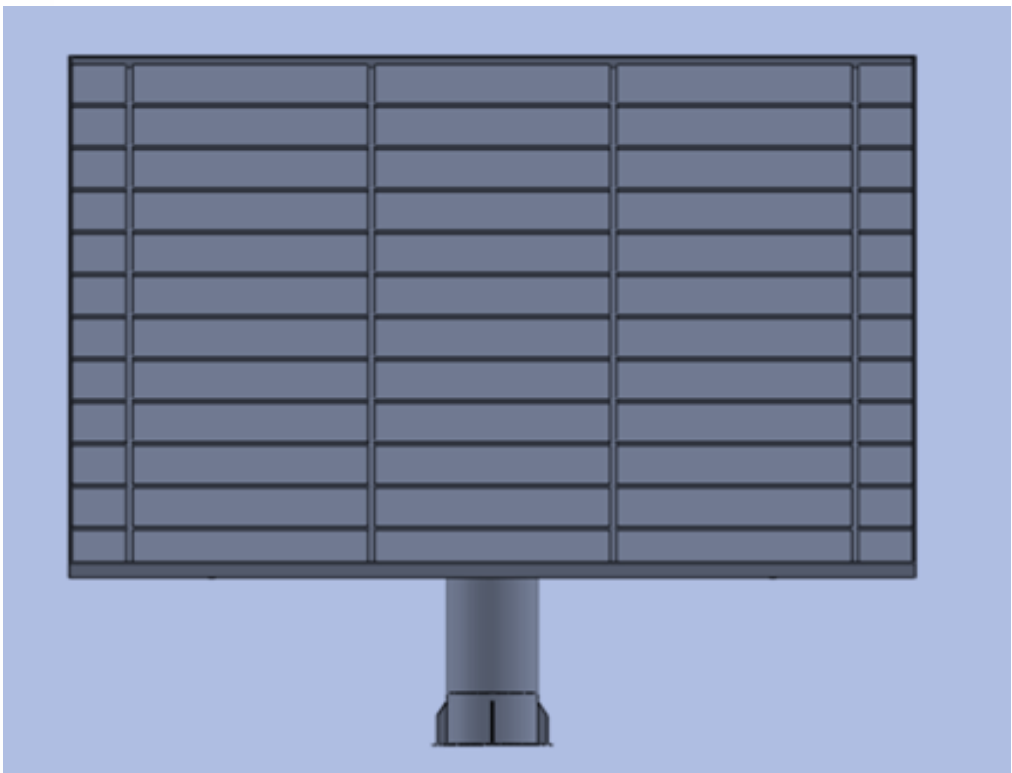
PROPOSED SOLUTIONS & DESIGNS



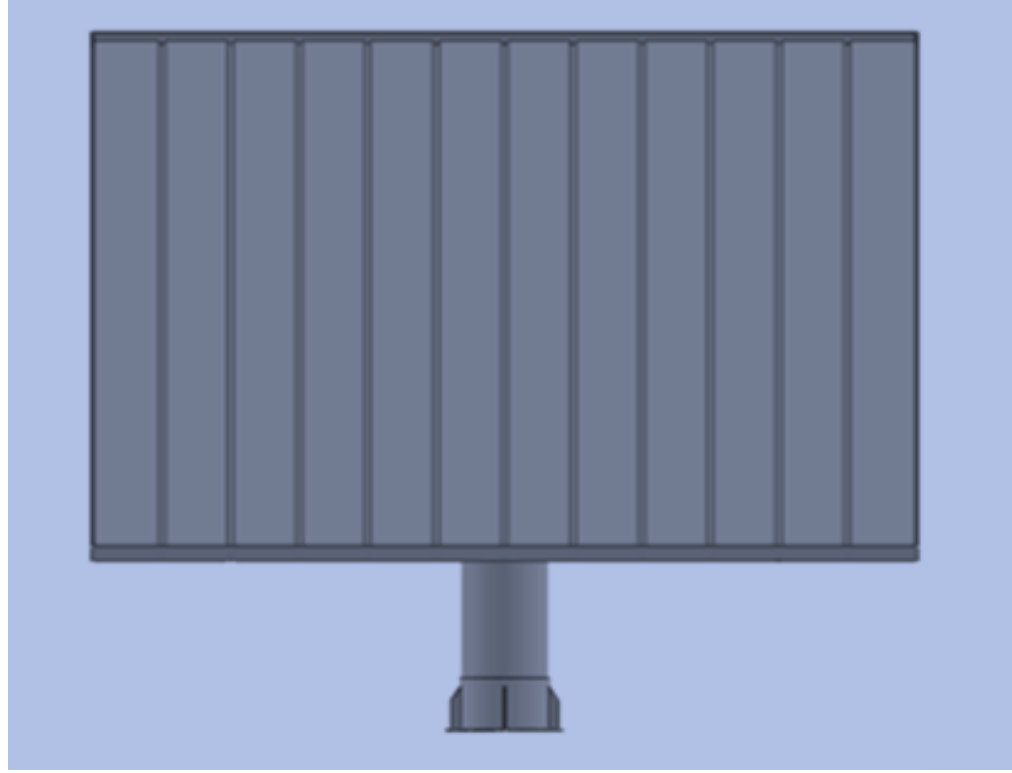
MSS



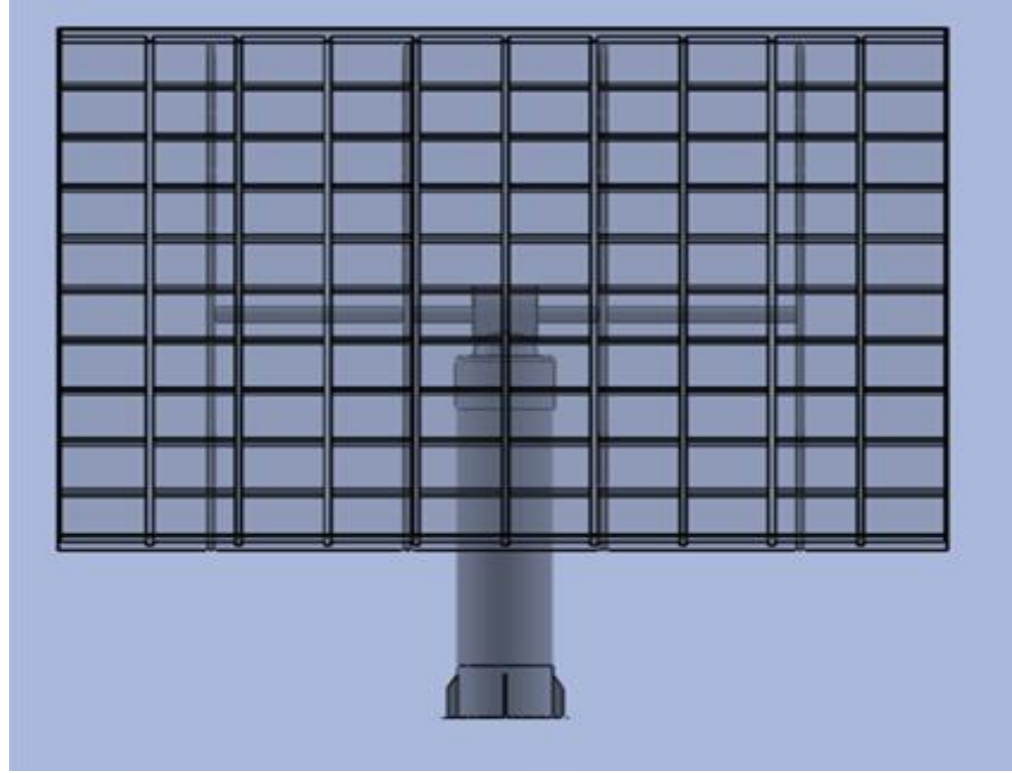
BYJ



DHH



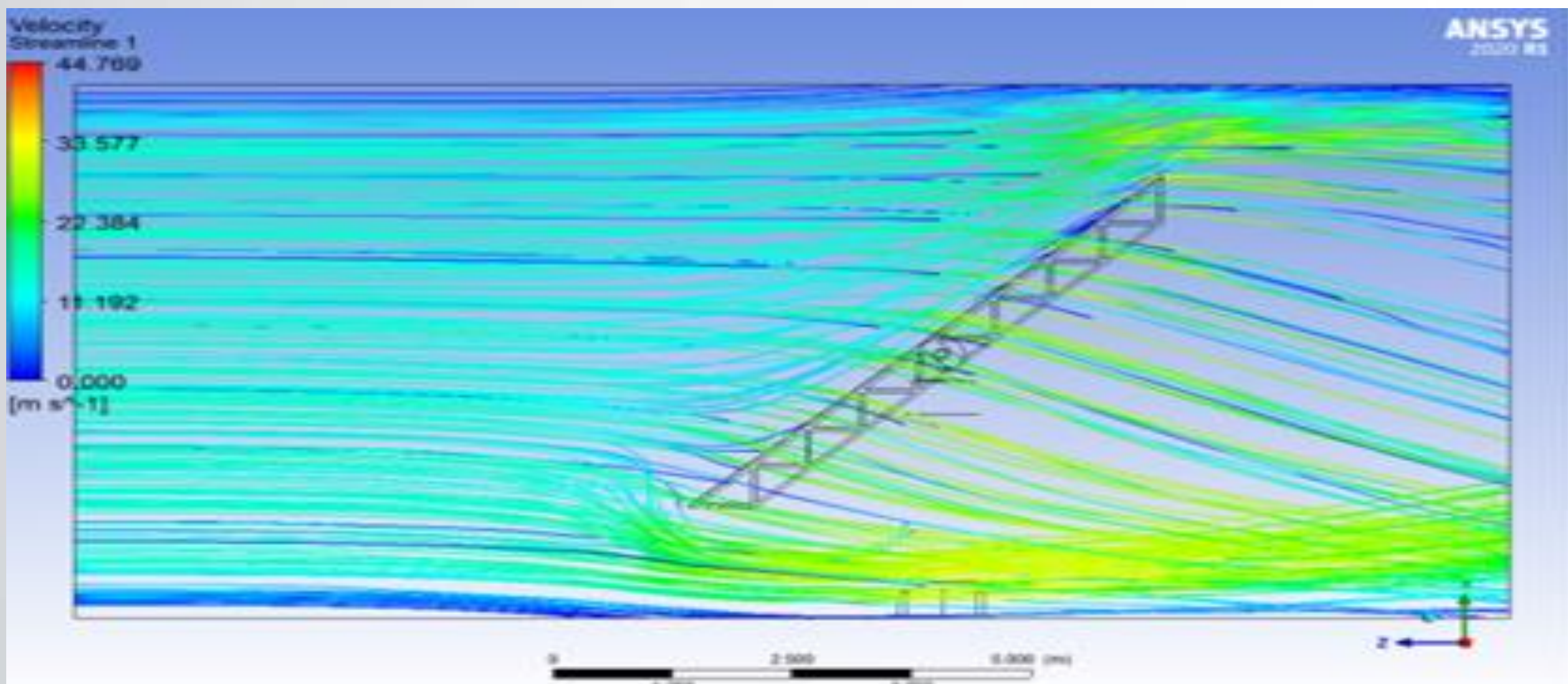
MKA



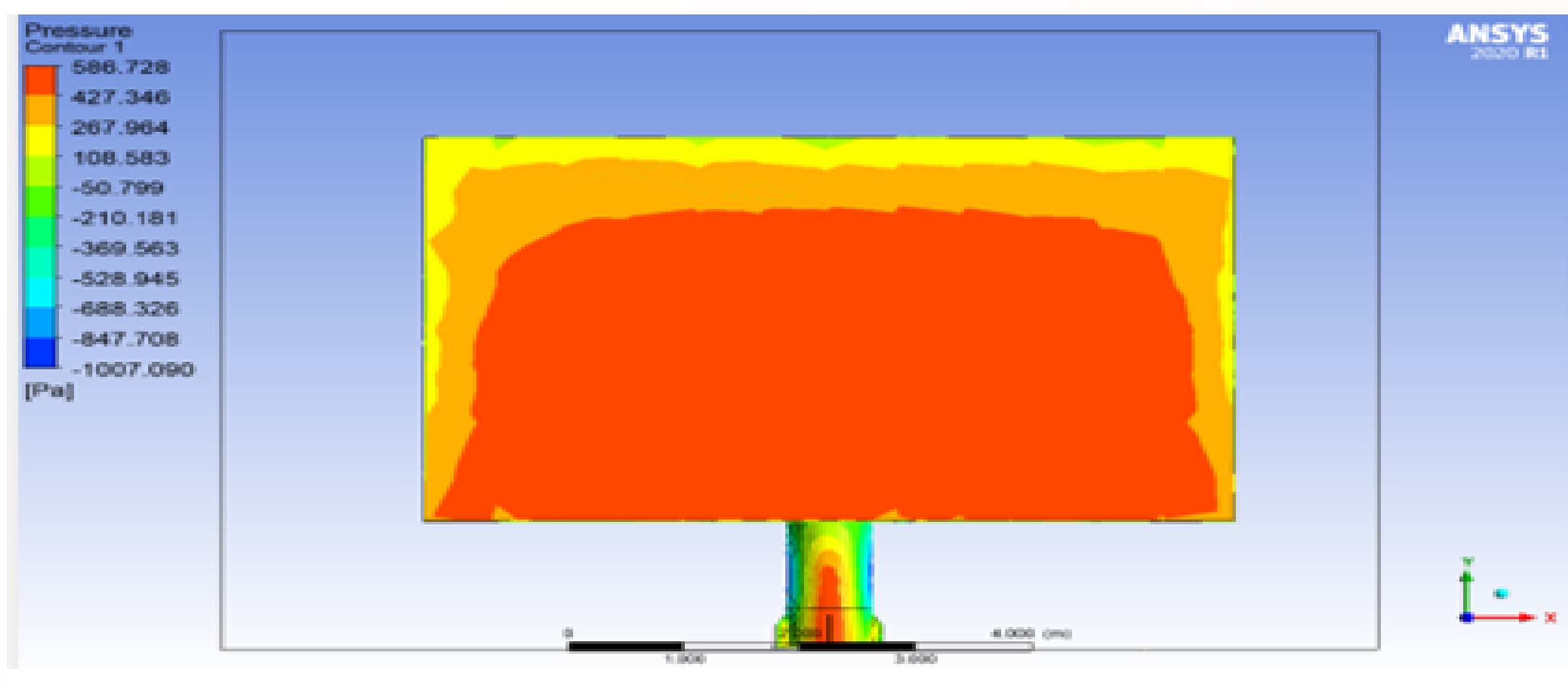
LAK

IMPLEMENTATION AND RESULTS

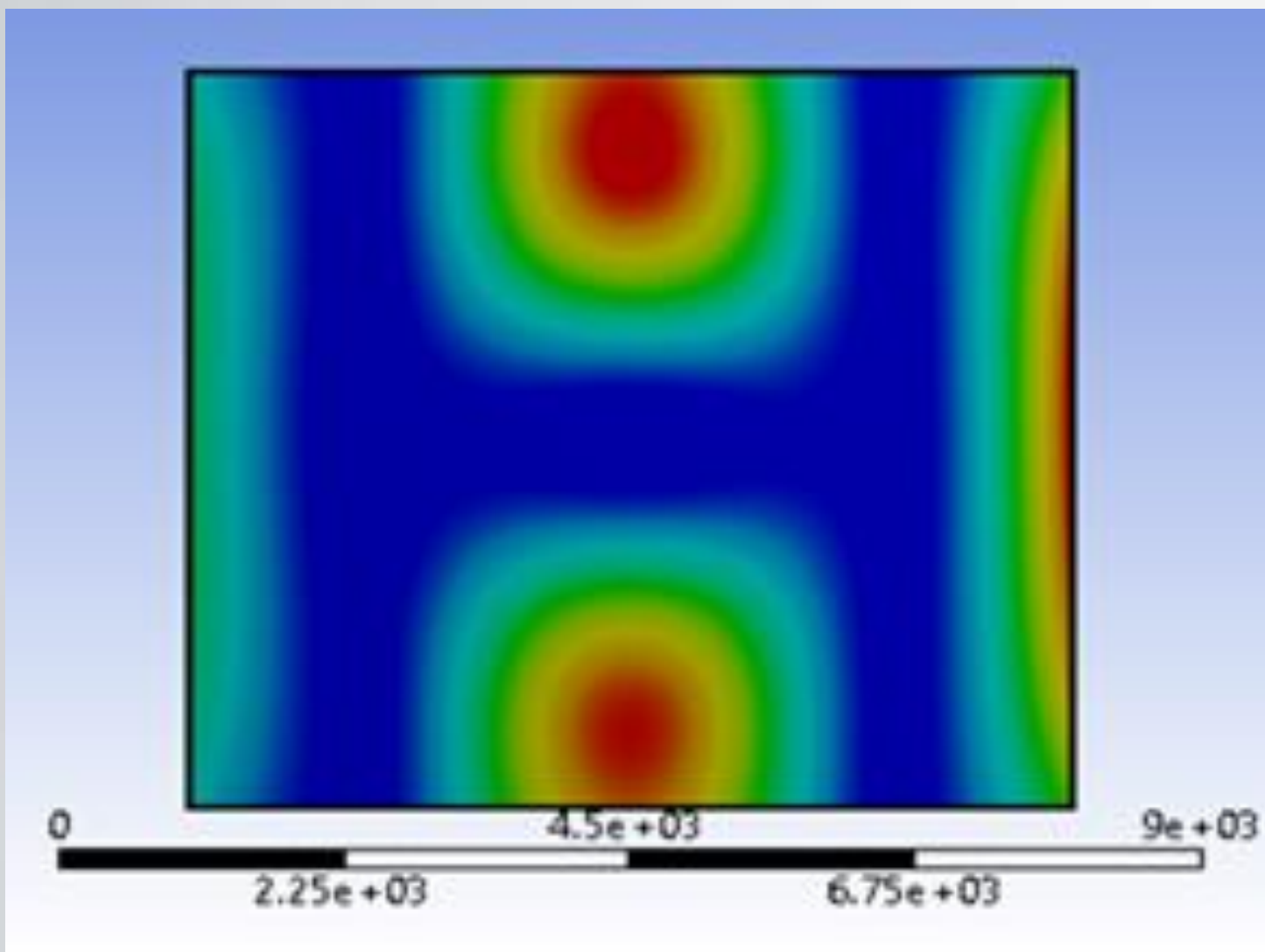
Design	MSS	BYJ	DHH	MKA	LAK
Modes on $\alpha=35$	Frequency [Hz]				
1	1.6622	2.1323	1.0158	1.8098	1.2614
2	1.17945	2.2023	1.577	1.9204	2.2028
3	2.3643	2.2945	3.01	2.0842	2.3641
4	2.3866	2.4466	3.3066	2.1588	2.4777
Stress [Mpa]	91.65	178	233	229	211
Mass [kg]	6347.5	4075.4	4166.3	5061	4290.1
FoS	2.62	1.35	1.03	1.05	1.14



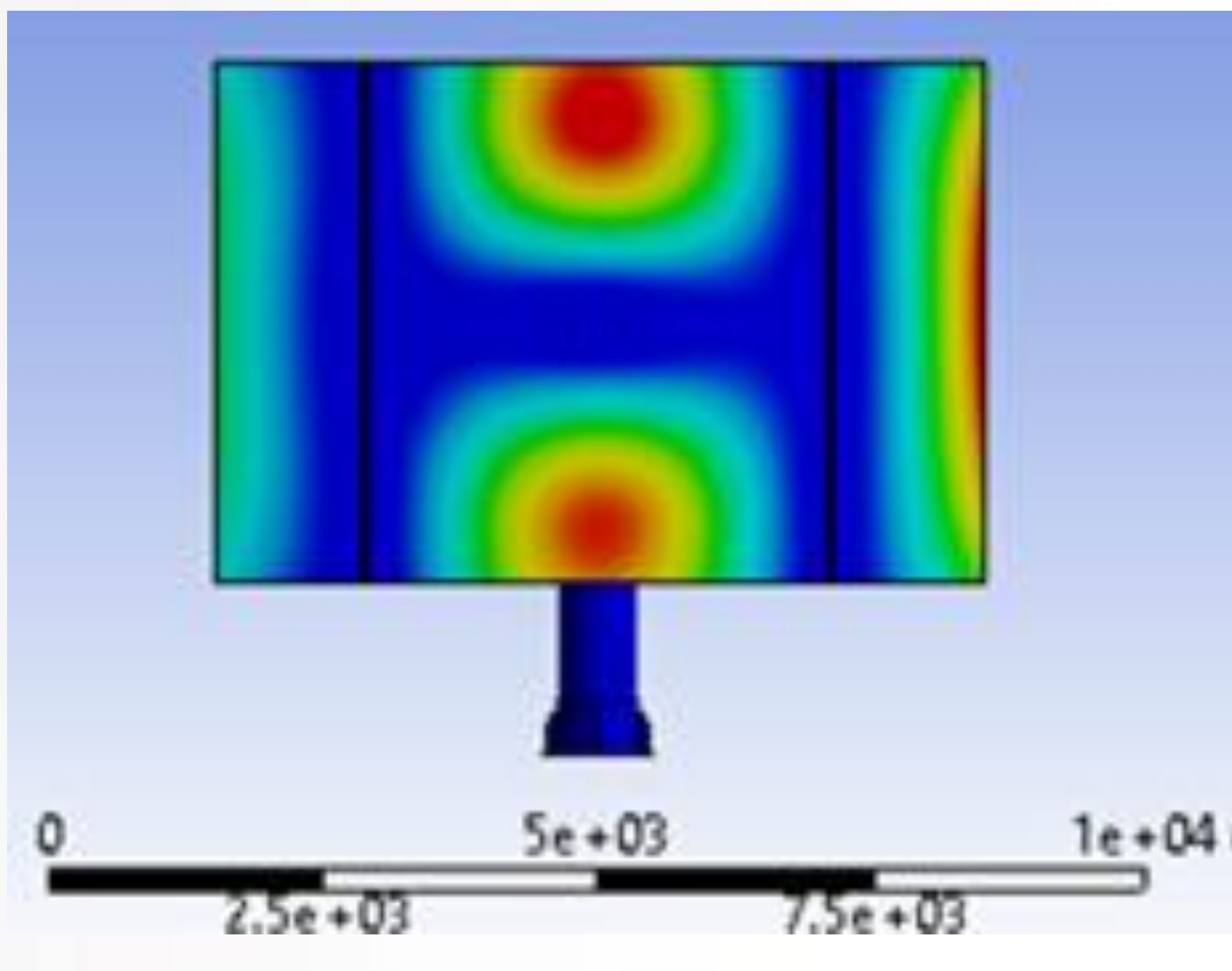
Velocity Streamline for BYJ



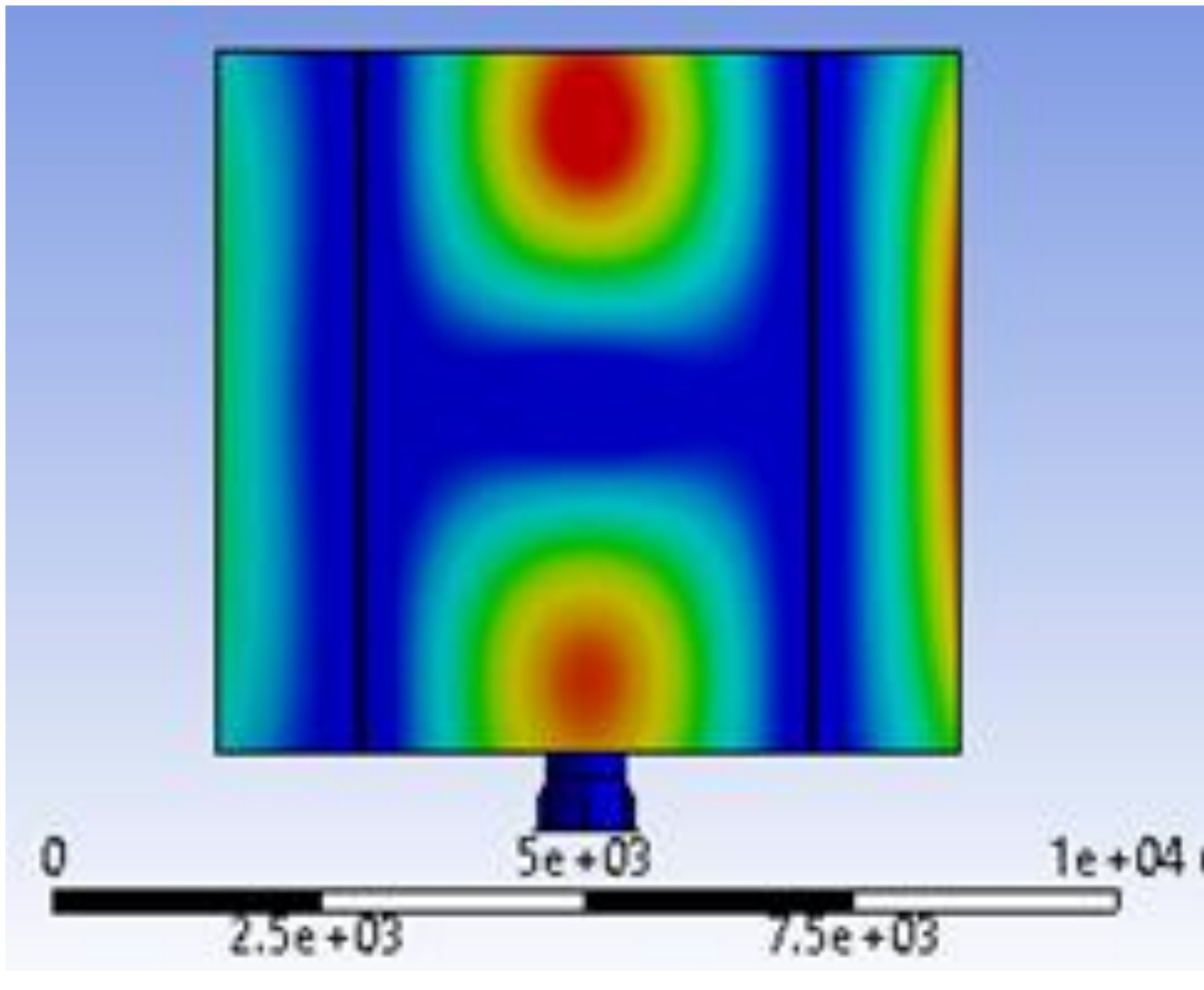
Pressure Contour for BYJ



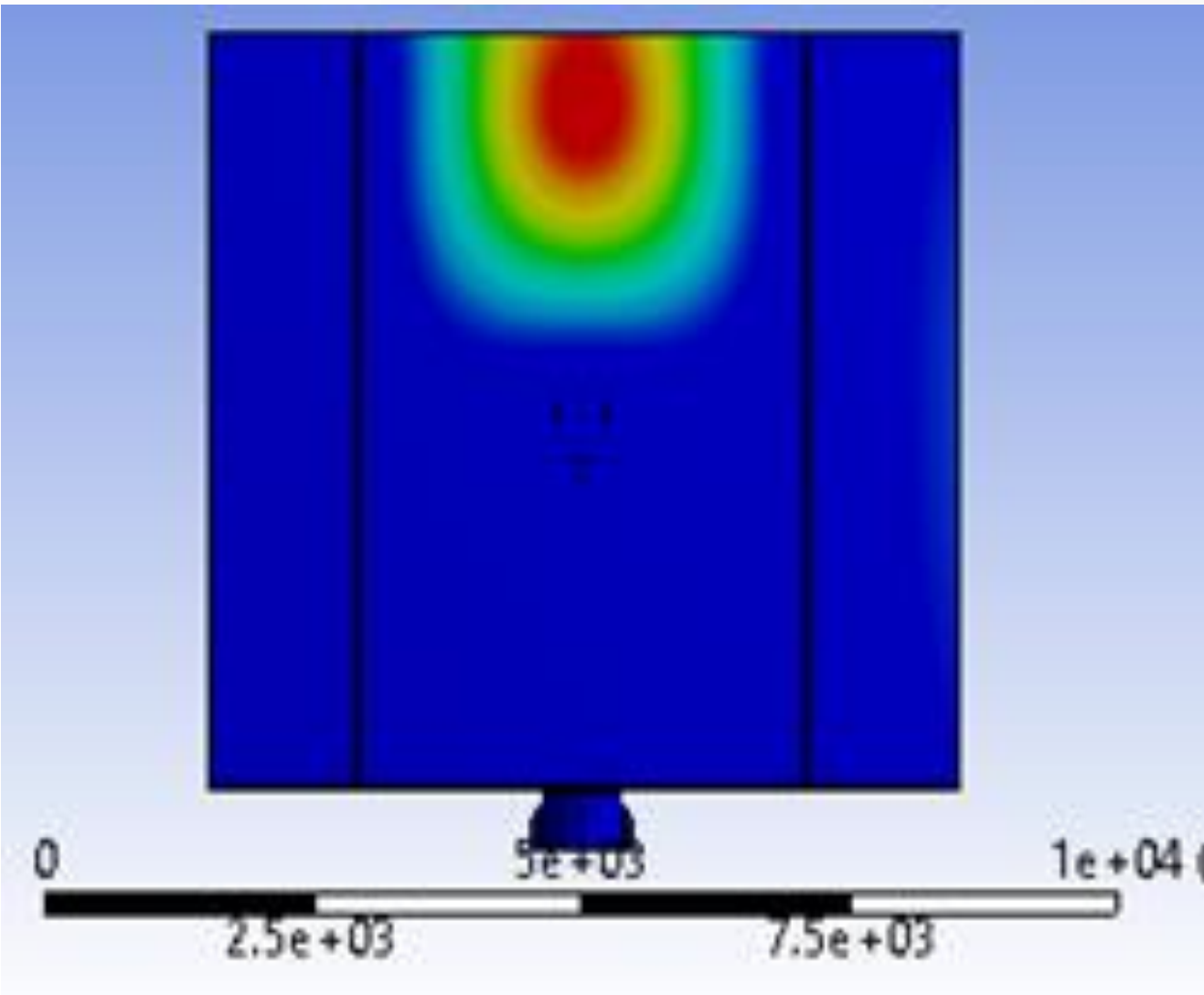
$\alpha = 0$, mode 1, (2.1033 Hz)



$\alpha = 35$, mode 1, (2.0973 Hz)



$\alpha = 70$, mode 1, (2.0898 Hz)



$\alpha = 90$, mode 1, (2.4266 Hz)

CONCLUSION AND FUTURE WORK

The design and analysis process for the heliostat in CSP plants is crucial to ensure the safety and efficiency of the plant. By using the QFD approach, developing CAD designs, simulating designs using ANSYS Fluent, and performing modal analysis, the optimum design was identified. There are many potential areas of future work to pursue that utilize more advanced simulation tools, materials, sensing technologies, and control strategies. An ideal heliostat design would be self-monitoring, self-optimizing, and easily adaptable - achieving the goals of high reliability, longevity and efficiency.

REFERENCES:

Gong, B., Wang, Z., Li, Z., Zang, C., & Wu, Z. (2013). Fluctuating wind pressure characteristics of heliostats. Renewable energy, 50, 307-316.
Björkman, N. (2014). Heliostat Design (Master's thesis, Indian Institute of Science, Indian). Retrieved from <https://www.diva-portal.org/smash/get/diva2:769446/FULLTEXT01.pdf>
Pfahl, A., Brucks, A., & Holze, C. (2014). Wind load reduction for light-weight heliostats. Energy Procedia, 49, 193-200.